

Willian Vieira PhD

Data scientist and ML engineer with a PhD in quantitative ecology. I build reproducible machine learning pipelines on noisy, real-world data — from Bayesian and ML modelling to the production infrastructure that makes models traceable, monitored, and safe to deploy.

DATA SCIENCE & ENGINEER EXPERIENCE

2025
|
2024
(1 yr 9 m)

• Data Analyst

Habitat, Montreal, Canada

Architected end-to-end R/Python ML pipelines to process and model large-scale, noisy real-world data for production analytics | Built metadata-driven ETL and provenance workflows to make datasets, models, and analyses traceable, reproducible, and reliable | Designed cloud data infrastructure on Azure for streaming heterogeneous data and optimizing feature retrieval for ML training and inference | Pushed ML models into production, supporting client-facing products with validation checks and monitoring | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, and DevOps best practices

2024
|
2017

• PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Built Bayesian hierarchical models to forecast forest dynamics from noisy, biased, continental-scale climate and field-inventory datasets | Developed reproducible HPC workflows () to run thousands of simulations in parallel for model evaluation, uncertainty propagation, and large-scale inference | Built end-to-end data pipelines to ingest, clean, harmonize, and model heterogeneous datasets across North America | Developed open-source software libraries (, ) for spatial demographic modelling with version control, automated testing, and reproducible workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (, )

2022
|
2020
(part-time)

• Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **data pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

2024
|
2017

• PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016
|
2015

• Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

 *Modelling the dispersion of weed species in agricultural landscapes*

2015
|
2010

• BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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 [LinkedIn/WillVieira](#)

 [GitHub/WillVieira](#)

 [Publications](#)

SKILLS

Machine Learning & Statistics:

Bayesian inference · Hierarchical models · Model evaluation · Error analysis · Uncertainty quantification · Forecasting · Feature engineering · Calibration

Programming & Database:

Python · R · SQL · PyTorch · Stan · DuckDB · Bash

Data Engineering & Cloud:

ETL/ELT · Production ML pipelines · Docker · Azure · AWS · Apache Arrow · HPC

DevOps & Reproducibility:

Git · CI/CD · Unix · Unit testing · Reproducible pipelines · Automated validation · Make · Jupyter

OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught programming, statistics, and ecology to 250+ students across biology, engineering, and graduate programs · Published **5 papers & preprints** with ~100 citations · Translated stakeholder needs into technical analyses, reproducible workflows, and clear analytical outputs

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

CV source code hosted on 

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